

Natural language statement

Domain

GroundTruth

Theorem 1.3.2. Let (X, μ) be a sigma-finite measure space, let (Y, ν) be another measure space, and let $0 < p_0 < p_1 \leq \infty$. Let T be a sublinear operator defined on $L^{p_0}(X) + L^{p_1}(X) = \{f_0 + f_1 : f_j \in L^{p_j}(X), j = 0, 1\}$ and taking values in the space of measurable functions on Y. Assume that there exist $A_0, A_1 < \infty$ such that $\ T(f)\ _{L^{p_0}(\nu)} \leq A_0 \ f\ _{L^{p_0}(X)}$ for all $f \in L^{p_0}(X)$, and $\ T(f)\ _{L^{p_1}(\nu)} \leq A_1 \ f\ _{L^{p_1}(X)}$ for all $f \in L^{p_1}(X)$. Then for all $p_0 < p < p_1$ and for all $f \in L^p(X)$ we have the estimate $\ T(f)\ _{L^p(Y)} \leq A \ f\ _{L^p(X)}$, where $A = 2[p/(p - p_0) + p/(p_1 - p)]^{1/p} A_0^{p/(p_0 - p)} A_1^{p/(p_1 - p)}$.	Fourier analysis	<pre>def IsSublinearOperator {X : Type u} {Y : Type v} (T : (X -> C) -> Y -> C) : Prop := T 0 = 0 /\ (forall f g x, T (f + g) x <= T f x + T g x) /\ forall c : C, forall f x, T (c * f) x = c * T f x def HasWeakType {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C) (p : R) (C : R>=0inf) : Prop := forall f : X -> C, MemLp f (ENNReal.ofReal p) mu -> forall alpha : R, 0 < alpha -> nu {y ENNReal.ofReal alpha < T f y e} <= ENNReal.rpow (C * eLpNorm f (ENNReal.ofReal p) mu / ENNReal.ofReal alpha) p def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C) (p q : R>=0inf) (C : R>=0inf) : Prop := forall f : X -> C, MemLp f p mu -> MemLp (T f) q nu /\ eLpNorm (T f) q nu <= C * eLpNorm f p mu theorem grafakos_1_3_2_marcinkiewicz_interpolation {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) [SigmaFinite mu] (T : (X -> C) -> Y -> C) {p0 p1 p : R} {A0 A1 : R>=0inf} (hp : 0 < p0 /\ p0 < p /\ p < p1) (hA0 : A0 < inf) (hA1 : A1 < inf) (hT : IsSublinearOperator T) (h0 : HasWeakType mu nu T p0 A0) (h1 : HasWeakType mu nu T p1 A1) : HasStrongType mu nu T (ENNReal.ofReal p) (ENNReal.ofReal p) (2 * ENNReal.rpow (ENNReal.ofReal (p / (p - p0) + p / (p1 - p))) (1 / p) * ENNReal.rpow A0 ((p0 / p) * ((p1 - p) / (p1 - p0))) * ENNReal.rpow A1 ((p1 / p) * ((p - p0) / (p1 - p0)))) := by sorry</pre>
Theorem 1.3.4. Let (X, μ) and (Y, ν) be two sigma-finite measure spaces. Let T be a linear operator defined on the set of all finitely simple functions on X and taking values in the set of measurable functions on Y. Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ and assume that $\ T(f)\ _{L^{q_0}(\nu)} \leq M_0 \ f\ _{L^{p_0}(X)}$ and $\ T(f)\ _{L^{q_1}(\nu)} \leq M_1 \ f\ _{L^{p_1}(X)}$ for all finitely simple functions f on X. Then for all $0 < \theta < 1$ we have $\ T(f)\ _{L^q(\nu)} \leq M_0^{1-\theta} M_1^\theta \ f\ _{L^p(X)}$ for all finitely simple functions f on X, where $1/p = (1-\theta)/p_0 + \theta/p_1$ and $1/q = (1-\theta)/q_0 + \theta/q_1$. Consequently, when $p < \infty$, by density, T has a unique bounded extension from $L^p(X, \mu)$ to $L^q(Y, \nu)$ when p and q are as above.	Fourier analysis	<pre>def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C) (p q : R>=0inf) (C : R>=0inf) : Prop := forall f : X -> C, MemLp f p mu -> MemLp (T f) q nu /\ eLpNorm (T f) q nu <= C * eLpNorm f p mu theorem grafakos_1_3_4_riesz_thorin_interpolation {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> l[C] (Y -> C)) {p0 p1 q0 q1 p q : R>=0inf} {theta : R} {M0 M1 : R>=0inf} (htheta : 0 < theta /\ theta < 1) (hexponents : 1 <= p0 /\ 1 <= p1 /\ 1 <= q0 /\ 1 <= q1) (hM0 : M0 < inf) (hM1 : M1 < inf) (hp : p^(1-theta) = ENNReal.ofReal (1 - theta) * p0^(1-theta) + ENNReal.ofReal theta * p1^(1-theta)) (hq : q^(1-theta) = ENNReal.ofReal (1 - theta) * q0^(1-theta) + ENNReal.ofReal theta * q1^(1-theta)) (h0 : HasStrongType mu nu T p0 q0 M0) (h1 : HasStrongType mu nu T p1 q1 M1) : HasStrongType mu nu T p q (ENNReal.rpow M0 (1 - theta) * ENNReal.rpow M1 theta) := by sorry</pre>
Theorem 2.1.6. The uncentered and centered Hardy-Littlewood maximal operators M and M^c map $L^1(\mathbb{R}^n)$ to $L^{1,\infty}(\mathbb{R}^n)$ with constant at most 3^n and also $L^p(\mathbb{R}^n)$ to $L^p(\mathbb{R}^n)$ for $1 < p < \infty$ with constant at most $3^{n/p} p(p-1)^{-1}$. For any $f \in L^1(\mathbb{R}^n)$ we also have $M(f) ^\alpha \leq (3^n/\alpha) \int M(f) ^\alpha dy$.	Fourier analysis	<pre>noncomputable def hardyLittlewoodMaximal (n : N) (f : EuclideanSpace R (Fin n) -> C) (x : EuclideanSpace R (Fin n)) : R>=0inf := y : EuclideanSpace R (Fin n), r : {r : R // 0 < r}, (_ : x in ball y r), (integral^~ z in ball y r, f z e) / volume (ball y r) noncomputable def hardyLittlewoodCenteredMaximal (n : N) (f : EuclideanSpace R (Fin n) -> C) (x : EuclideanSpace R (Fin n)) :</pre>

Natural language statement

Domain

GroundTruth

		<pre> R>=0inf := r : {r : R // 0 < r}, (integral^~ z in ball x r, f z e) / volume (ball x r) theorem grafakos_2_1_6_hardy_littlewood_maximal {n : N} : let operators := ({hardyLittlewoodMaximal n, hardyLittlewoodCenteredMaximal n} : Set ((EuclideanSpace R (Fin n) -> C) -> EuclideanSpace R (Fin n) -> R>=0inf)) (forall M in operators, forall f : EuclideanSpace R (Fin n) -> C, MemLp f 1 volume -> forall alpha : R, 0 < alpha -> volume {x ENNReal.ofReal alpha < M f x} <= ENNReal.ofReal (3 ^ n / alpha) * integral^~ x in {x ENNReal.ofReal alpha < M f x}, f x e) /\ forall M in operators, forall p : R, 1 < p -> forall f : EuclideanSpace R (Fin n) -> C, MemLp f (ENNReal.ofReal p) volume -> ENNReal.rpow (integral^~ x, ENNReal.rpow (M f x) p) (1 / p) <= ENNReal.ofReal (3 ^ ((n : R) / p) * p / (p - 1)) * eLpNorm f (ENNReal.ofReal p) volume := by sorry </pre>
Theorem 2.2.14. Given f, g, and h in $S(\mathbb{R}^n)$, we have (1) $\int_{\mathbb{R}^n} f(x)g(x) dx = \int_{\mathbb{R}^n} f(x)g(x) dx$; (2) (Fourier Inversion) $(fV) = f = (fV)$; (3) (Parseval's relation) $\int_{\mathbb{R}^n} f(x)h(x) dx = \int_{\mathbb{R}^n} f(x)h(x) dx$; (4) (Plancherel's identity) $\ f\ _{L^2} = \ f\ _{L^2}$ = $\ fV\ _{L^2}$; (5) $\int_{\mathbb{R}^n} f(x)h(x) dx = \int_{\mathbb{R}^n} f(x)hV(x) dx$.	Fourier analysis	<pre> theorem grafakos_2_2_14_fourier_identities_on_schwartz {n : N} (f g h : S(EuclideanSpace R (Fin n), C)) : ((integral x, f x * F g x) = integral x, F f x * g x) /\ F^~ (F f) = f /\ F (F^~ f) = f /\ (integral x, star (F f x) * F g x) = integral x, star (f x) * g x /\ eLpNorm (fun x => F f x) 2 volume = eLpNorm f 2 volume /\ eLpNorm (fun x => F^~ f x) 2 volume = eLpNorm f 2 volume /\ (integral x, F f x * h x) = integral x, f x * F h x := by sorry </pre>
Proposition 2.2.16 (Hausdorff-Young inequality). For every function f in $L_p(\mathbb{R}^n)$ we have the estimate $\ f\ _{L^p} \leq \ f\ _{L^p}$ whenever $1 \leq p \leq 2$.	Fourier analysis	<pre> theorem grafakos_2_2_16_hausdorff_young {n : N} {p : R} {f : EuclideanSpace R (Fin n) -> C} (hp : 1 <= p /\ p <= 2) (hf : MemLp f (ENNReal.ofReal p) volume) : let conjugateExponent : R>=0inf := if p = 1 then inf else ENNReal.ofReal (p / (p - 1)) MemLp (F f) conjugateExponent volume /\ eLpNorm (F f) conjugateExponent volume <= eLpNorm f (ENNReal.ofReal p) volume := by sorry </pre>
Theorem 3.2.8 (Poisson summation formula). Let f be a continuous function on $\mathbb{R}^n$ which satisfies $f(x) \leq C(1+ x)^{-n-\delta}$ for some $C, \delta > 0$ and whose Fourier transform $\hat{f}$ restricted on $\mathbb{Z}^n$ satisfies $\sum_{m \in \mathbb{Z}^n} \hat{f}(m) < \infty$. Then for all x in $\mathbb{R}^n$, $\sum_{m \in \mathbb{Z}^n} f(m)e^{2\pi i m \cdot x} = \sum_{k \in \mathbb{Z}^n} f(x+k)$, and in particular $\sum_{m \in \mathbb{Z}^n} f(m) = \sum_{k \in \mathbb{Z}^n} f(k)$.	Fourier analysis	<pre> theorem grafakos_3_2_8_poisson_summation {n : N} {f : EuclideanSpace R (Fin n) -> C} (hf : Continuous f /\ Integrable f /\ (exists C delta : R, 0 < C /\ 0 < delta /\ forall x, f x <= C * (1 + x) ^ (-(n : R) - delta)) /\ Summable fun m : Fin n -> Z => F f ((WithLp.toLp 2 fun i => (m i : R)))) : (forall x : EuclideanSpace R (Fin n), sum' m : Fin n -> Z, F f ((WithLp.toLp 2 fun i => (m i : R))) * Complex.exp (2 * Real.pi * Complex.I * (sum i, (m i : C) * (x i : C)))) = sum' k : Fin n -> Z, f (x + (WithLp.toLp 2 fun i => (k i : R)))) /\ (sum' m : Fin n -> Z, F f ((WithLp.toLp 2 fun i => (m i : R)))) = sum' k : Fin n -> Z, f ((WithLp.toLp 2 fun i => (k i : R))) := by sorry </pre>
Theorem 4.1.1. For $R > 0$ and m in $\mathbb{Z}^n$, let $a(m, R)$ be complex numbers such that (i) for every $R > 0$ there is a qR such that $a(m, R) = 0$ if $m > qR$; (ii) there is an $M_0 < \infty$ such that $a(m, R) \leq M_0$ for all m in $\mathbb{Z}^n$ and all $R > 0$; (iii) for each m in $\mathbb{Z}^n$, the limit of $a(m, R)$ exists as $R \rightarrow \infty$ and $\lim_{R \rightarrow \infty} a(m, R) = a_m$. Let $1 \leq p < \infty$. For f in $L_p(\mathbb{T}^n)$ and x in $\mathbb{T}^n$ define $S_R(f)(x) = \sum_{m \in \mathbb{Z}^n} a(m, R)f(m)e^{2\pi i m \cdot x}$, noting that the sum is well defined because of (i). Also, for h in $C_{inf}(\mathbb{T}^n)$ define $A(h)(x) = \sum_{m \in \mathbb{Z}^n} a_m h(m)e^{2\pi i m \cdot x}$. Then for all f in $L_p(\mathbb{T}^n)$ the sequence $S_R(f)$ converges in L_p as $R \rightarrow \infty$ if and only if there exists a constant $K < \infty$ such	Fourier analysis	<pre> def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C) (p q : R>=0inf) (C : R>=0inf) : Prop := forall f : X -> C, MemLp f p mu -> MemLp (T f) q nu /\ eLpNorm (T f) q nu <= C * eLpNorm f p mu noncomputable def torusCharacter {n : N} (m : Fin n -> Z) (x : Fin n -> AddCircle (1 : R)) : C := prod i, fourier (m i) (x i) </pre>

Natural language statement

Domain

GroundTruth

that $\sup_{R>0} \|S_R\|_{L^p \rightarrow L^p} \leq K$. Furthermore, if this holds, then for the same constant K we have $\sup_{\{h \in C_0^\infty, h \neq 0\}} \|A(h)\|_{L^p} / \|h\|_{L^p} \leq K$, and then A extends to a bounded operator A from $L^p(T^n)$ to itself; moreover, for every f in $L^p(T^n)$ we have that $S_R(f) \rightarrow A(f)$ in L^p as $R \rightarrow \infty$.

```
noncomputable def torusFourierCoefficient {n : ℕ}
  [MeasurableSpace (Fin n → AddCircle (1 : ℝ))]
  (mu : Measure (Fin n → AddCircle (1 : ℝ)))
  (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) (m : Fin n → ℝ) : ℝ :=
  integral x, star (torusCharacter m x) * f x dmu

theorem grafakos_4_1_1_torus_summability_uniform_boundedness
  {n : ℕ}
  (a : ℝ → (Fin n → ℝ) → ℝ) (alimit : (Fin n → ℝ) → ℝ)
  (hfinite : forall R, 0 < R → (Function.support (a R)).Finite)
  (hbounded : exists M : ℝ, 0 ≤ M ∧ forall R m, 0 < R → ||a R m|| ≤ M)
  (htendsto : forall m, Tendsto (fun R => a R m) atTop (nhds (alimit m)))
  {p : ℝ} (hp : 1 ≤ p) :
  let mu : Measure (Fin n → AddCircle (1 : ℝ)) := volume
  let S := fun R (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) =>
    (x : Fin n → AddCircle (1 : ℝ)) =>
      sum' m, a R m * torusFourierCoefficient mu f m * torusCharacter m x
  let A := fun (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) =>
    (x : Fin n → AddCircle (1 : ℝ)) =>
      sum' m, alimit m * torusFourierCoefficient mu f m * torusCharacter
        m x
  ((forall f, MemLp f (ENNReal.ofReal p) mu →
    Tendsto (fun R => eLpNorm (S R f - A f) (ENNReal.ofReal p) mu)
      atTop (nhds 0)) <=>
    exists C : ℝ → ℝ, C < inf ∧ forall R, 0 < R →
      HasStrongType mu mu (S R) (ENNReal.ofReal p) (ENNReal.ofReal p) C)
    ∧
  forall C : ℝ → ℝ, C < inf ∧ forall R, 0 < R →
    (forall R, 0 < R → HasStrongType mu mu (S R) (ENNReal.ofReal p)
      (ENNReal.ofReal p) C) →
      HasStrongType mu mu A (ENNReal.ofReal p) (ENNReal.ofReal p) C := by
  sorry
```

Theorem 4.3.15. For every $1 < p < \infty$ there exists a finite constant C_p such that for all f in $C_0^\infty(\mathbb{R})$ we have $\|C^{**}(f)\|_{L^p(\mathbb{R})} \leq C_p \|f\|_{L^p(\mathbb{R})}$, where $C^{**}(f)(x) = \sup_{R>0} |\int_{|x| \leq R} f(x) e^{2\pi i x} dx|$ is the Carleson operator.

Fourier
analysis

```
noncomputable def carlesonHuntMaximal (f : S(ℝ, ℝ)) (x : ℝ) : ℝ → ℝ :=
  R : {R : ℝ // 0 < R},
  ||integral xi in Set.Icc (-R.1) R.1,
    f xi * Complex.exp (2 * Real.pi * Complex.I * xi * x)||e

theorem grafakos_4_3_15_carleson_hunt_line {p : ℝ} (hp : 1 < p) :
  exists C : ℝ → ℝ, C < inf ∧ forall f : S(ℝ, ℝ),
    ENNReal.rpow (integral^~ x, ENNReal.rpow (carlesonHuntMaximal f x) p)
      (1 / p) <=
      C * eLpNorm (f : ℝ → ℝ) (ENNReal.ofReal p) volume := by
  sorry
```

Theorem 5.3.1. Let f in $L^1(\mathbb{R}^n)$ and $\alpha > 0$. Then there exist functions g and b on $\mathbb{R}^n$ such that $f = g + b$; $\|g\|_{L^1} \leq \|f\|_{L^1}$ and $\|g\|_{L^\infty} \leq 2^n \alpha$; $b = \sum_j b_j$ where each b_j is supported in a dyadic cube Q_j , the cubes are disjoint, $\int_{Q_j} b_j(x) dx = 0$, $\|b_j\|_{L^1} \leq 2^{n+1} \alpha \|f\|_{L^1}$, and $\sum_j \|b_j\|_{L^1} \leq \alpha^{-1} \|f\|_{L^1}$.

Fourier
analysis

```
structure DyadicCube (n : ℕ) where
  scale : ℝ
  corner : Fin n → ℝ

def DyadicCube.carrier {n : ℕ} (Q : DyadicCube n) : Set (EuclideanSpace ℝ (Fin n)) :=
  {x | forall i,
    (Q.corner i : ℝ) * 2 ^ Q.scale ≤ x i ∧
    x i < (Q.corner i + 1 : ℝ) * 2 ^ Q.scale}

theorem grafakos_5_3_1_calderon_zygmund_decomposition
  {n : ℕ} {f : EuclideanSpace ℝ (Fin n) → ℝ} {alpha : ℝ}
  (hf : MemLp f 1 volume) (halpha : 0 < alpha) :
  exists g b : EuclideanSpace ℝ (Fin n) → ℝ,
  exists (Q : ℕ → DyadicCube n) (bad : ℕ → EuclideanSpace ℝ (Fin n)
    → ℝ),
  f = g + b ∧ HasSum bad b ∧ eLpNorm g 1 volume ≤ eLpNorm f 1
    volume ∧
  eLpNorm g inf volume ≤ ENNReal.ofReal (2 ^ n * alpha) ∧
  (Pairwise fun i j => Disjoint (Q i).carrier (Q j).carrier) ∧
  (forall j, Function.support (bad j) subset (Q j).carrier) ∧
  (forall j, integral x, bad j x = 0) ∧
  (forall j, eLpNorm (bad j) 1 volume ≤
    ENNReal.ofReal (2 ^ (n + 1) * alpha) * volume (Q j).carrier) ∧
  sum' j, volume (Q j).carrier ≤ eLpNorm f 1 volume /
    ENNReal.ofReal alpha := by
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Natural language statement

Domain

GroundTruth

		sorry
Theorem 5.6.6. For $1 < p, r < \infty$ the Hardy-Littlewood maximal function M satisfies the vector-valued inequalities $\ (\sum_j M(f_j) ^r)^{1/r}\ _{1,\infty} \leq C_n(1+(r-1)^{-1})\ (\sum_j f_j ^r)^{1/r}\ _1$ and $\ (\sum_j M(f_j) ^r)^{1/r}\ _p \leq C_n c(p,r)\ (\sum_j f_j ^r)^{1/r}\ _p$.	Fourier analysis	<pre>noncomputable def hardyLittlewoodMaximal (n : ℕ) (f : EuclideanSpace ℝ (Fin n) → ℝ) (x : EuclideanSpace ℝ (Fin n)) : ℝ := 0 y : EuclideanSpace ℝ (Fin n), r : {r : ℝ // 0 < r}, (λ z : x in ball y r), (integral^ z in ball y r, f z e) / volume (ball y r) theorem grafakos_5_6_6_vector_valued_maximal {n : ℕ} {p r : ℝ} (hp : 1 < p) (hr : 1 < r) : let ellNorm := fun (f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ) x => ENNReal.rpow (sum' j, ENNReal.rpow f j x e r) (1 / r) let maximalNorm := fun (f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ) x => ENNReal.rpow (sum' j, ENNReal.rpow (hardyLittlewoodMaximal n (f j) x) r) (1 / r) exists Cn Cp : ℝ := 0 Cn < inf /\ Cp < inf /\ (forall f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ, forall alpha : ℝ, 0 < alpha -> volume {x ENNReal.ofReal alpha < maximalNorm f x} <= Cn * ENNReal.ofReal (1 + 1 / (r - 1)) / ENNReal.ofReal alpha * ENNReal.rpow (integral^ x, ellNorm f x) 1) /\ forall f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ, ENNReal.rpow (integral^ x, ENNReal.rpow (maximalNorm f x) p) (1 / p) <= Cp * ENNReal.rpow (integral^ x, ENNReal.rpow (ellNorm f x) p) (1 / p) := by sorry</pre>